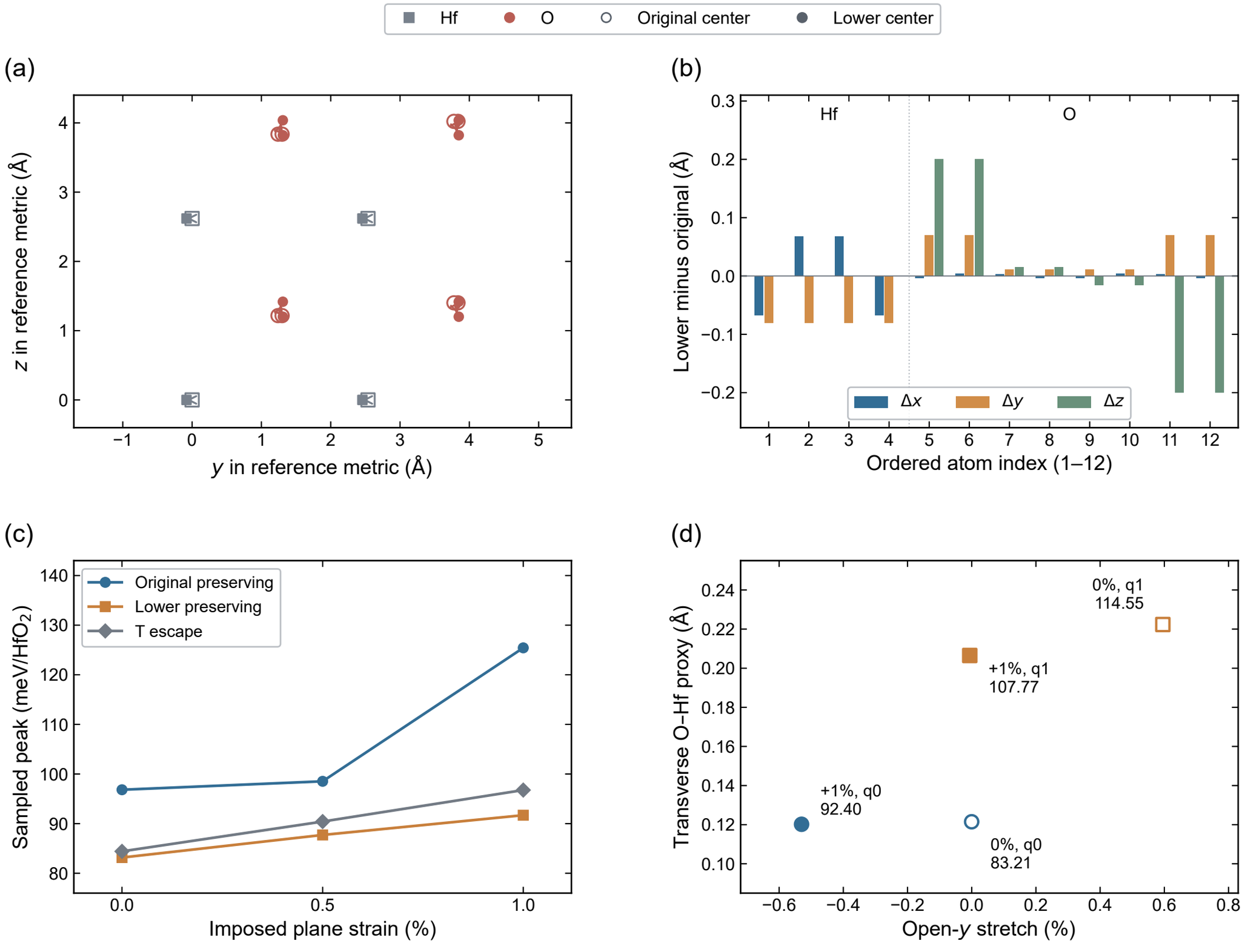
Hf
O
Original center
Lower center
(a)
z in reference metric (Å)
y in reference metric (Å)
(b)
Hf
O
Lower minus original (Å)
Δx
Δy
Δz
Ordered atom index (1–12)
(c)
Original preserving
Lower preserving
T escape
Sampled peak (meV/HfO2)
Imposed plane strain (%)
(d)
0%, q1
114.55
+1%, q1
107.77
+1%, q0
92.40
0%, q0
83.21
Transverse O–Hf proxy (Å)
Open-y stretch (%)